Homework 5

```r
library(tidyverse)
library(mdsr)
library(dbplyr)
library(DBI)
```

```r
# connect to the database which lives on a remote server maintained by
#   St. Olaf's IT department

library(RMariaDB)
con <- dbConnect(
  MariaDB(), host = "mdb.stolaf.edu",
  user = "ruser", password = "ruserpass",
  dbname = "flight_data"
)
```

On Your Own - Adapting 164 Code - HW

1. Summarize carriers flying to MSP by number of flights and proportion that are cancelled (assuming that a missing arrival time indicates a cancelled flight). [This was #4 in 17_longer_pipelines.Rmd.]

First duplicate the output above, then check trends in 2023 across all origins. Here are a few hints:

- use flightdata instead of flights_nyc13
- remember that flights_nyc13 only contained 2013 and 3 NYC origin airports (EWR, JFK, LGA)
- is.na can be replaced with CASE WHEN ArrTime IS NULL THEN 1 ELSE 0 END or with CASE WHEN cancelled = 1 THEN 1 ELSE 0 END
- CASE WHEN can also be used replace fct_collapse

```
SELECT
  origin,
  CASE
    WHEN Reporting_Airline IN ('DL', '9E') THEN 'Delta +'
    WHEN Reporting_Airline IN ('AA', 'MQ') THEN 'American +'
    ELSE 'United +' END AS carrier,
    COUNT(*) AS num_flights,
  SUM(CASE WHEN ArrTime IS NULL THEN 1 ELSE 0 END) AS num_cancelled,
  AVG(CASE WHEN cancelled = 1 THEN 1 ELSE 0 END) AS prop_cancelled
FROM flightdata
WHERE Year = 2023 AND dest = 'MSP' AND origin IN ('LGA', 'EWR', 'JFK')
GROUP BY origin, carrier
LIMIT 0, 20;
```

Table 1: 6 records

origin	carrier	num_flights	num_cancelled	prop_cancelled
EWR	Delta +	1308	23	0.0176
EWR	United +	1073	30	0.0280
JFK	Delta +	1049	20	0.0191
JFK	United +	147	5	0.0340
LGA	Delta +	1729	35	0.0202
LGA	United +	632	12	0.0190

2. Plot number of flights vs. proportion cancelled for every origin-destination pair (assuming that a missing arrival time indicates a cancelled flight). [This was #7 in 17_longer_pipelines.Rmd.]

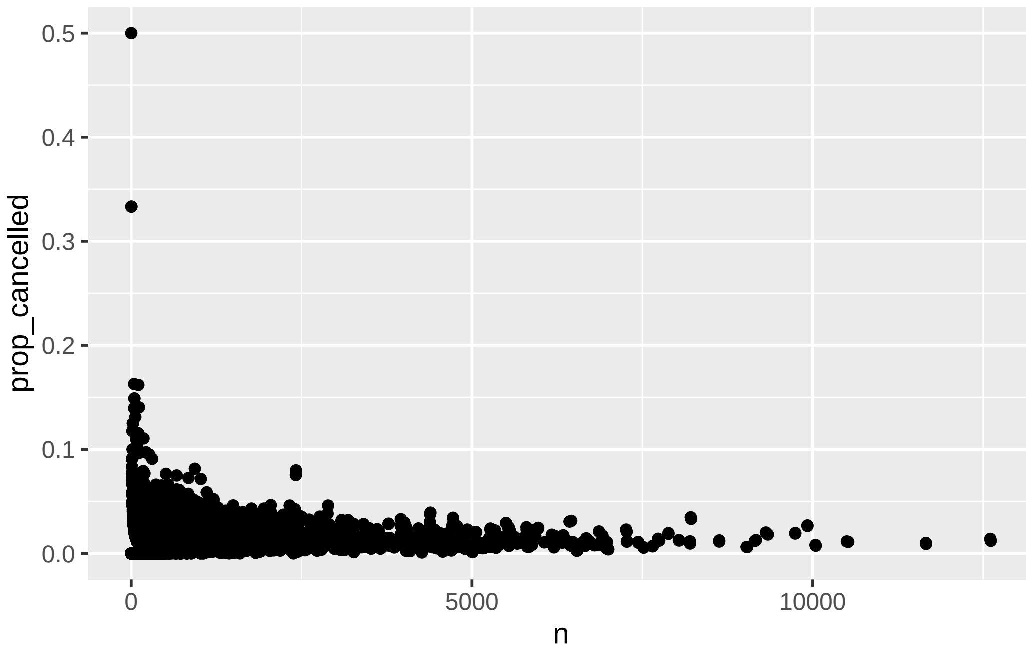
First duplicate the plot above for 2023 data, then check trends across all origins. Do all of the data wrangling in SQL. Here are a few hints:

- use flightdata instead of flights_nyc13
- remember that flights_nyc13 only contained 2013 and 3 NYC origin airports (EWR, JFK, LGA)
- use an `sql` chunk and an `r` chunk
- include `connection =` and `output.var =` in your sql chunk header (this doesn't seem to work with dbGetQuery()…)

```
my_flights_data <- dbGetQuery(con, '
SELECT origin, dest,
  SUM(1) AS n,
```

```
  AVG(CASE WHEN cancelled = 1 THEN 1 ELSE 0 END) AS prop_cancelled
FROM flightdata
WHERE Year = 2023
GROUP BY origin, dest
HAVING prop_cancelled < 1;
')

ggplot(my_flights_data, aes(n, prop_cancelled)) +
  geom_point()
```



3. Produce a table of weighted plane age by carrier, where weights are based on number of flights per plane. [This was #6 in 26_more_joins.Rmd.]

First duplicate the output above for 2023, then check trends across all origins. Do all of the data wrangling in SQL. Here are a few hints:

- use flightdata instead of flights_nyc13
- remember that flights_nyc13 only contained 2013 and 3 NYC origin airports (EWR, JFK, LGA)
- you'll have to merge the flights dataset with the planes dataset
- you can use DISTINCT inside a COUNT()
- investigate SQL clauses for calculating a standard deviation
- you cannot use a derived variable inside a summary clause in SELECT

For bonus points, also merge the airlines dataset and include the name of each carrier and not just the abbreviation!

```sql
SELECT
  a.name,
  f.Reporting_Airline AS carrierID,
  COUNT(DISTINCT f.TAIL_NUMBER) AS unique_planes,
  AVG(2023 - p.year) AS mean_weighted_age,
  STDDEV_SAMP(2023 - p.year) AS sd_weighted_age
FROM flightdata AS f
JOIN planes AS p ON f.TAIL_NUMBER = p.tailnum
JOIN airlines AS a ON f.Reporting_Airline = a.carrier
GROUP BY carrier
ORDER BY mean_weighted_age ASC
LIMIT 0, 10;
```

Table 2: Displaying records 1 - 10

name	carrierID	unique_planes	mean_weighted_age	sd_weighted_age
Frontier Airlines Inc.	F9	156	5.1488	2.875346
Spirit Air Lines	NK	213	6.8567	3.858105
Horizon Air	QX	47	7.8606	4.268428
PSA Airlines Inc.	OH	170	11.3136	5.682408
Alaska Airlines Inc.	AS	251	11.4766	6.752782
Envoy Air	MQ	272	11.8303	7.188317
Mesa Airlines Inc.	YV	137	12.2227	5.601301
SkyWest Airlines Inc.	OO	581	12.5643	6.851667
Southwest Airlines Co.	WN	820	12.7597	6.419219
Republic Airline	YX	204	12.7657	4.189681